Vamshi Krishna Konyala | Data Analyst

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SUMMARY

Data Analyst with hands-on experience in the **financial services, healthcare, and IT sectors**, specializing in **data analysis, reporting, and visualization**. Skilled in leveraging **SQL, Python, Power BI, Tableau, and advanced Excel** to extract insights from complex datasets and enable **data-driven decision-making**. Adept at building interactive dashboards, optimizing data pipelines, and ensuring **data integrity and compliance** in regulated industries. Experienced in collaborating with cross-functional teams across global environments, applying **statistical methods, forecasting, and machine learning techniques** to solve business problems. Proactive, detail-oriented, and adaptable professional with a proven ability to bridge **business needs and technical solutions** while driving measurable impact.

TECHNICAL SKILLS

Programming & Querying: SQL, Python, R, DAX, Power Query
Data Analysis & Statistics: Data Wrangling, Data Cleansing, Data Validation, Hypothesis Testing, A/B Testing, KPI Frameworks, Descriptive & Inferential Statistics, Forecasting, Anomaly Detection
Visualization & Reporting: Power BI (RLS, DAX, Power Query), Tableau (LOD, Storytelling Dashboards), Qlik Sense, MS Excel (PivotTables, Solver, VBA Macros)
Databases & Data Warehousing: MySQL, PostgreSQL, MongoDB, Snowflake, AWS Redshift, Azure Synapse, GCP BigQuery, ETL (Airflow, Alteryx, Talend, SSIS, Informatica)
Machine Learning & Advanced Analytics: Regression (Linear, Logistic, Ridge, Lasso), Decision Trees, Random Forests, Gradient Boosting (XGBoost, LightGBM, CatBoost), Clustering (K-Means, DBSCAN, Hierarchical), Time Series (ARIMA, SARIMA, Prophet), NLP (Sentiment Analysis, Text Analytics basics), Anomaly Detection
Cloud & Big Data Platforms: AWS (S3, RDS, EC2, Redshift, Lambda, SageMaker), Azure (Data Factory, Synapse, Databricks), GCP (BigQuery, Dataflow)
Tools, Methodologies & Collaboration: Git, GitHub, GitLab CI/CD, Docker, Jenkins, MLflow, Agile (Scrum, Kanban), Waterfall, CRISP-DM, SDLC, Lean Six Sigma, JIRA, Confluence, MS Visio, Lucidchart, SharePoint, MS Teams

EXPERIENCE

Data Analyst | Fidelity Information Services (FIS Global), USA **Sep 2025 – Present**

- Gathered, collected, and **cleaned large and complex datasets** from multiple internal and external sources to support **financial services operations**. Ensured **data accuracy, integrity, and compliance** with regulatory standards.
- Fine-tuned and sustained interactive **Power BI/Tableau dashboards** to track KPIs and operational metrics, enabling management to make **data-driven strategic decisions**, resulting in a **20% faster decision-making process**.
- Utilized statistical methods and quantitative techniques to identify trends, patterns, and anomalies in financial data, supporting **risk assessment, forecasting, and performance optimization**, which **decreased financial discrepancies by 15%**.
- Refined and streamlined complex **SQL queries** for data extraction, transformation, and analysis; documented processes and identified automation opportunities, **reducing manual reporting effort by 25%**.

Data Analyst Intern | Centene Health, MA, USA **Jan 2025 – Apr 2025**

- Extracted, collected, and cleaned large datasets from multiple sources, including **healthcare claims, provider data, and member information**. Ensured **data accuracy and integrity** to support analytics projects and reliable business insights.
- Assisted in developing and maintaining interactive Power BI dashboards and visual reports. Tracked key performance indicators (KPIs) and operational metrics, enabling management to make data-driven decisions efficiently, **leading to faster reporting cycles by 15%**.
- Collaborated with **cross-functional teams**, including IT and business analysts, **to improve existing data processes**, identify opportunities for automation, and enhance overall internal customer experience, **reducing process inefficiencies by 15%**.
- Provided analytical support for various business operations and projects, translating complex datasets into **actionable insights**. Contributed to initiatives aimed at optimizing managed care performance and operational efficiency, **helping improve operational KPIs by 10–12%**.
- Documented technical requirements, project findings, and data processes meticulously. Ensured compliance with federal and state reporting requirements, maintaining accurate records for audits and future reference, **ensuring 100% regulatory compliance**.

Data Analyst | LTI Mindtree, India **Apr 2021 – Dec 2022**

- Gathered, cleaned, and analyzed large and complex datasets from multiple internal and external sources. Identified **trends, patterns, and anomalies**, enabling **data-driven decisions** and providing actionable insights to improve business performance.
- Collaborated with cross-functional stakeholders in finance, marketing, and operations to understand reporting and analytical requirements. Translated these requirements into actionable insights, KPIs, and recommendations for strategic decision-making, **supporting faster business decisions by 20%**.
- Planned, expanded, and preserved interactive Power BI/Tableau dashboards and visual reports. Presented complex datasets in a clear and intuitive manner, enabling stakeholders to quickly track performance metrics, trends, and key indicators, **reducing report preparation time by 25%**.
- Developed, optimized, and maintained SQL queries for data extraction, transformation, and reporting. Conducted thorough data validation, reconciliation, and integrity checks to ensure accurate and reliable datasets for business use, **cutting data errors by 15%**.
- Used Python and advanced Excel (pivot tables, VLOOKUP, macros) for statistical analysis, scenario modeling, and forecasting. Streamlined reporting processes, improved analytical efficiency, and decreased manual effort for the team, **saving 10+ hours per week**.

PROJECTS

Automated Retail Price Forecasting with MLOps

- Planned and deployed a **retail price prediction pipeline** implementing **MLOps principles** using **Apache Airflow**, automating all stages including **data ingestion, preprocessing, model training, and deployment** to ensure continuous, reliable operation.
- Processed **500K+ retail transactions** from multiple sources, performing **feature engineering, cleaning, and transformation** to capture seasonal and demand patterns relevant for price forecasting.
- Experimented with multiple regression models including **SGDRegressor, Decision Tree Regressor, and KNN**, and performed **hyperparameter tuning using MLflow**, improving model accuracy by **25%** and reducing **MAPE error by 30%**.
- Deployed the solution on **Google Cloud Platform (GCP)** with **Slack-based real-time monitoring and computerized alerts**, ensuring smooth pipeline operation and immediate issue resolution for production-grade reliability.

AI-Powered Breast Cancer Detection

- Built a **deep learning model** using **TensorFlow/Keras fully connected neural networks** to predict breast cancer, achieving **95% classification accuracy**, significantly outperforming baseline machine learning methods.
- Applied advanced **data preprocessing techniques** including **feature scaling, PCA for dimensionality reduction, and class balancing**, enhancing the model's generalization across diverse patient datasets.
- Leveraged the **MIAS mammographic dataset**, validating model performance using **cross-validation, confusion matrix analysis, and ROC metrics**, ensuring robust and clinically reliable predictions.
- Incorporated **explainable AI methods (SHAP, LIME)** to interpret model predictions, providing transparency and enabling **healthcare professionals to trust AI-assisted decision-making**.

Intelligent Bicycle Sharing Optimization System

- Fine-tuned an **AI-driven optimization system** for a bicycle sharing network, analyzing **10K+ rental transactions** to optimize **bike distribution, station placement, and rental pricing** across urban locations.
- Applied **machine learning models** such as **Random Forest and Gradient Boosting** to predict demand, achieving **20% higher bike availability** during peak hours and reducing **trip duration prediction error by 15%**.
- Designed a **hybrid database architecture** combining **MySQL for structured data and MongoDB for unstructured logs**, enabling **real-time monitoring, analytics, and demand forecasting** for efficient operations.
- Integrated **seasonal demand forecasting models** to predict usage trends and support **data-driven city transport planning**, enhancing overall operational efficiency and user satisfaction.

EDUCATION

- **Master of Science in Data Analytics Engineering** **Jan 2023 – May 2025**
  Northeastern University, Boston, MA, USA
- **Bachelor of Technology in Mechanical Engineering** **Jun 2018 – Jun 2022**
  Gandhi Institute of Technology and Management (G.I.T.A.M), Visakhapatnam, Andhra Pradesh, India

CERTIFICATIONS

Google Data Analytics Professional Course

- Attained Coursera Certification by completing coursework, projects, and assessments with 95% score
- Demonstrated expertise in eight sub-courses of the Google Data Professional curriculum